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On the Justice of Earnings: A New Specification of the Justice Evaluation Function¹

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This paper describes a new specification of the justice evaluation function. It predicts precisely and unambiguously the kind of injustice (i.e., underreward or overreward) and the degree of injustice associated with given departures from perfect justice. The new justice evaluation function was inducted from extensive analyses of survey data on the perceived justice or injustice of earnings. It is directly generalizable to cover all socially distributed goods, and hence I propose it as a candidate for a universal Law of Justice Evaluation in distributional matters. Finally, I suggest that such a Law of Justice Evaluation performs one of the three tasks essential to a scientific theory of distributive justice.

This paper reports a new, empirically supported specification of the justice evaluation function, one that simply and so far accurately predicts the precise magnitude of injustice associated with measured departures from perfect justice in the distribution of earned income, and one that can be extended to cover all distributional matters.

Recent empirical work on the perceived justice of earnings (Jasso and Rossi 1977) explores the content of and the degree of consensus about the principles of justice governing earnings, and estimates just earnings amounts for incumbents of a variety of both occupational and other classifications in contemporary American society. Jasso and Rossi's (1977) research design uses judgments about the fairness or relative unfairness of earnings amounts randomly attached to fictitious persons and families in order to infer the just earnings for such units. Their major findings are interrespondent agreement on the principles of justice and a just earnings distribution characterized by small variance, highly restricted range, and monetary returns not only to education and occupation but also to sex and family claims.

In this paper I explore the justice evaluation function by assessing the magnitudes of injustice associated with given deviations from perfect justice. Jasso and Rossi (1977) recognize the existence of a justice evaluation

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1398 *AJS* Volume 83 Number 6

variable and integrate it within their research design. They represent it as an Aristotelian dimension spanning the real-number line, with zero, the origin of the line, chosen to represent the point of perfect justice, the positive and negative segments to represent judgments of overreward and underreward, respectively, and the absolute value of the justice evaluation to represent the magnitude of injustice. They report quadratic justice evaluation functions and use them to estimate just earnings.

I suggest here that finding the “true” shape of the justice evaluation function is an imperative task. It is not enough to know the parameters of the just earnings distribution. We live in a world that rarely realizes congruence between actual earnings and just earnings. The major questions of social life are not limited to questions about “what is just.” Rather, they include questions about how unjust is this or that aspect of social reality and about the responses of individuals and groups to the observed degree of injustice.

Social scientists have long recognized the importance of the response to injustice (e.g., Homans 1974, 1976; Berger et al. 1972) and have suggested that judging oneself overpaid produces guilt and judging oneself underpaid produces anger. They have not, however, had much to say about responses to judgments about injustice received by others. Moreover, the empirical record shows negligible support for the guilt hypothesis (see Adams and Freedman 1976, pp. 45–46).

I wish to suggest that, before there can be any precision in the determination of responses to the two kinds of injustice, there must be precise measures of the magnitudes of injustice and that unambiguous specification of the justice evaluation function is the immediate, essential work before us.

This paper begins that work. It reports a new quantitative specification of the sense of justice and injustice concerning the distribution of earned income. The specification I believe to be correct was inducted from extensive reanalyses of the survey data gathered by Jasso and Rossi (1977). Its many merits make it immediately a candidate for a universal Law of Justice Evaluation in all distributional matters.

I begin with a look at the requirements for an adequate justice evaluation function and a careful examination of previous formulations. I next describe the data set used in the present research and report the analyses that led to induction of the lawlike relation governing the sense of justice and injustice. Finally, I discuss the justice evaluation function and its implications.

SPECIFYING THE JUSTICE EVALUATION: A TASK IN THE STUDY OF DISTRIBUTIVE JUSTICE

The most ancient version of the problem of distributive justice searches for the one just distribution of resources; it seeks to know “what is just” and is

therefore a problem in normative philosophy. The scientist's version of the problem seeks, rather, to answer three questions: (1) What do people think is just? (2) Do individuals distinguish degrees of injustice, and, if so, how are these related to departures from perfect justice? (3) What is the behavioral response to injustice? This is a problem in descriptive ethics or, in more modern terminology, in scientific explanation.

This paper focuses on the second question. It asks about the sense of injustice that accompanies deviations from the just order. Are there magnitudes of injustice? Is underreward thought to be more unjust than overreward? What is the precise relation between the just reward and the justice evaluation? Not until these questions are answered can the search for responses to injustice yield quantified, precise results.

I now examine the requirements for an adequate specification of the justice evaluation function and then review the existing formulations.

Requirements for a Law of Justice Evaluation

We begin by formalizing the discussion so far, and doing so at the level of the individual and for the distribution of one good. Since the data to be analyzed in this paper concern earnings, we will use earnings as the good in question (while remembering that the formal structure can apply to any other good).

The following terms are used:

1. *Observer*.—This is the person making the judgments about the justice or injustice of the earnings received by any earner, including him or herself.

2. *Just earnings*.—This is the global amount the observer considers just for a given earner. The term is devoid of ontological content; it only represents the observer's judgment, whatever it may be. For example, the observer may judge that just earnings for a movie star are \$25,000 a year and for the president of a corporation \$100,000 or that just earnings for *all* earners are \$8,000 a year.

3. *Actual earnings*.—This is the actual yearly earnings received by any given earner.

4. *Justice*.—This is the condition when an individual's just earnings exactly equal his or her actual earnings.

5. *Injustice*.—This is the condition in which an individual's just earnings and actual earnings are unequal. Injustice is of two kinds: (a) unjust underreward—the case when just earnings exceed actual earnings and (b) unjust overreward—the case when actual earnings exceed just earnings.

Now, what are the requirements for a Law of Justice Evaluation? What would such a law look like? It can readily be seen that it is easy enough to posit a specification for justice: justice exists whenever actual earnings equal just earnings; in fact, this is true by definition.

What is not so easy is to posit a specification of injustice. A genuine Law of Justice Evaluation must offer a single simple formula that, given two terms only, actual earnings and just earnings, accurately predicts every conceivable evaluation, including perfect justice and degrees of underreward and overreward. That is, our task is to specify the shape of the function (see Homans 1967, p. 18). An adequate justice evaluation function must predict exactly how the justice evaluation will change when the two terms, actual earnings and just earnings, change.

The value of the justice evaluation must be meaningful, independently of the good under consideration. That is, justice and injustice must be expressed in some meaningful, albeit nonfundamental, justice units, such that a given scale value always means the same thing, say, extreme injustice of the overreward kind or perfect justice.

Thus, we have two tasks: (1) to quantify justice and injustice and (2) to specify precisely the function that determines justice and injustice. A proposition that satisfies both these conditions will be regarded as a suitable candidate for a universal Law of Justice Evaluation.

Notice that such a law leaves entirely open the question of what comes to be regarded as just in the first place, a point which I return to later in the paper.

Formulas for the Justice Evaluation Function in the Literature

In this section I review the major works in distributive justice and examine the specifications of the justice evaluation function contained in them.

The work of Homans.—Homans (1974, 1976) followed Aristotle closely and suggested that justice is the result of local comparisons between the reward/investment ratios of individuals. Perfect justice is defined as equality of these ratios across persons. Injustice is defined as inequality of these ratios.

Stated formally, Homans's position, for the case of two individuals, A and B, is

$$\text{justice} \equiv \left[\frac{\text{A's reward}}{\text{A's investment}} = \frac{\text{B's reward}}{\text{B's investment}} \right],$$

which is equivalent to the following two expressions,

$$\text{justice} \equiv \left[\frac{\text{A's reward}}{\text{A's investment}} - \frac{\text{B's reward}}{\text{B's investment}} = 0 \right],$$

$$\text{justice} \equiv [(\text{reward}_A/\text{investment}_A)/(\text{reward}_B/\text{investment}_B) = 1].$$

The specification of justice made by Homans is unambiguous. Whichever of the equivalent expressions given above we use, the results are the same. Perfect justice is defined as zero difference between the two ratios or, in the alternative, as unity when we take the ratio of the two ratios.

However, the Homans proposition has two major weaknesses; both have

been pointed out by Berger et al. (1972), and both indicate its inadequacy as a justice evaluation function. First, it cannot specify the kind of injustice that results when the ratios are not equal; we have no way of knowing whether one person is underrewarded or the other overrewarded. Second, the Homans formulation cannot cope with the possibility that both A and B are unjustly rewarded (i.e., both equally underrewarded or equally overrewarded). Thus, the Homans proposition is of no use in the specification of the sense of injustice and may even be downright wrong in the prediction of justice.

That is, if we express Homans's proposition as a justice evaluation function,

$$\text{justice evaluation} = \frac{\text{A's reward}}{\text{A's investment}} - \frac{\text{B's reward}}{\text{B's investment}},$$

we see that equality of the ratios always produces a value of zero, but one which cannot be uniformly interpreted as justice, and that inequality of the ratios produces a number that not only varies with the position of A's and B's terms but also cannot be consistently interpreted as one or another kind of injustice.

And yet there remains an intuitive appeal in the Homans formulation. The notion of a ratio strikes responsive chords, as evidenced by the many scientists who have followed the pattern set by Homans, among them Adams (1965) and Waltster, Berscheid, and Walster (1976). We shall have occasion to return to this matter later in the paper.

The work of Berger, Zelditch, Anderson, and Cohen.—The model proposed by Berger et al. (1972) defines a situation as unjust if a real-world actor's level of a goal object differs from the level of the goal object received by his or her counterpart in a "referential structure." Berger et al. intuited an intrinsic aspect of the justice evaluation process when they posited a disembodied (Platonic) image of fairness. They observed correctly that the distinction between overreward and underreward is visible only in the presence of such a referential structure.

We can formalize their thinking as follows (for the sake of convenience, I use the term "reward" as shorthand for "level of a goal-object"):

$$\text{justice} \equiv [(\text{actual reward} - \text{just reward}) = 0].$$

Their specification of an evaluation of justice is both unambiguous and correct. But what about injustice? We now state the proposition of Berger et al. as a justice evaluation function:

$$\text{justice evaluation} = \text{actual reward} - \text{just reward}.$$

If the justice evaluation is zero, the condition of perfect justice obtains, and we can infer that Berger et al. would use zero to represent the point of justice. If the actual reward exceeds the just reward, the justice evaluation

assumes a positive value, and we can infer that they would use the positive segment of the real-number line to represent unjust overreward. If the just reward exceeds the actual reward, the justice evaluation assumes a negative value, and, likewise, we can infer that they would use the negative segment to represent underreward.

Now this is a wonderful achievement. Berger et al. have posited a single relation between two terms that predicts outcomes (i.e., scores for individuals) spanning the full range of the human sense of justice and injustice.

But the proposition of Berger et al. has a major weakness. Whenever the actual reward and the just reward are not equal, that is, whenever the justice evaluation process results in the sense of injustice, the value of the justice evaluation appears in the reward's units and not in meaningful justice units. That is, what should be a magnitude of injustice is, instead, a positive or negative magnitude of the good whose distribution is under evaluation. If, say, actual earnings are \$10,000 and just earnings are \$8,000, the justice evaluation equals \$2,000.

As noted earlier, the justice evaluation must be in justice units, units that are meaningful independently of the particular reward or good under consideration.

Moreover, there is an additional weakness in the formula of Berger et al. Whenever the absolute differences between actual earnings and just earnings are equal, the "injustice," albeit in wrong units, is equal. And this outcome appears to violate the human experience that deficiency is felt to be more unjust than a comparable excess.

And yet, the proposition of Berger et al. also has great intuitive appeal. I recognize the fidelity to common experience contained in the referential comparisons. And I see much merit in the conceptualization of injustice as some sort of difference between an actual reward and a just reward. We shall return to their formulation later in the paper.

Variants of the classical propositions.—In this section we will examine two formulations, one proposed as part of theory and descended from Homans (via Adams), the second inducted from empirical research and related to the formula of Berger et al.

a) The work of Walster, Berscheid, and Walster: Walster et al. (1976, pp. 2–4) offer a specification of justice founded on the Homans and Adams ratios (note: they use the terms "equity" and "inequity" as synonyms for "justice" and "injustice"):

We define an "equitable relationship" to exist when . . . all participants are receiving equal relative outcomes. . . . [W]e have chosen the following formula for calculating whether or not a relationship is equitable:

$$\frac{O_A - I_A}{(|I_A|)k_A} = \frac{O_B - I_B}{(|I_B|)k_B}$$

where O designates a scrutineer's perception of A's and B's outcomes, I designates his perception of A's and B's inputs, and $|I|$ designates the *absolute value* of A's and B's inputs (i.e., the perceived value of A's and B's inputs, disregarding sign). The exponents k_A and k_B take on the value $+1$ or -1 , depending on the sign of A and B's inputs and A and B's gains (Outcomes $-$ Inputs). [$k_A = \text{sign}(I_A) \times \text{sign}(O_A - I_A)$ and $k_B = \text{sign}(I_B) \times \text{sign}(O_B - I_B)$.] A participant's relative outcomes will be zero if his outcomes equal his inputs. His relative outcomes will be positive if his $O > I$, and negative if his $O < I$. Thus, the sign and the magnitude of this measure indicates how "profitable" the relationship has been to each of the participants.

The formula of Walster et al. has strengths and weaknesses. It makes significant progress toward a formula that will produce justice evaluation scores for individuals. Walster et al. do this by showing that each participant has a relative outcomes score of his or her own and that this score can take on values that span the real-number line.

On the other hand, the formula has the same fundamental weaknesses of the Homans ratios. First, it cannot deal adequately with situations in which both (or all) parties are unjustly rewarded. Since they have defined equity as equality of relative outcomes, they still have to call it equity when the participants have either negative or positive equal relative outcomes scores. A second weakness is that it is unduly complicated, as the ad hoc signs of the exponents indicate.

Moreover, even as it travels toward individuals' justice scores, the formula is plagued by the same two problems that befell the scores of Berger et al. First, the scores are in outcome/input units and not in justice (or equity) units (as evidenced by the fact that equity exists whenever the relative outcomes are equal, thereby eliminating the possibility of congruence between a justice scale and the relative outcome dimension). Second, equal differences between outcomes and inputs produce equal absolute magnitudes on the individuals' relative outcomes scores, so the formula again fails to reflect the common opinion that deficiency is more unjust than excess.

b) The work of Jasso and Rossi: In their analyses of the earnings data, Jasso and Rossi (1977) found empirical support for a model remarkably similar to that proposed by Berger et al. Using conventional techniques, they regressed the observed justice evaluations on the attributes of the fictitious units and fit a model which suggests that the justice evaluation process involves three analytic steps. First, the value, in justice units, of actual earnings is calculated. Second, the value, in justice units, of the set of justly perceived claims to earnings is calculated. Third, the justice value of the sum of the claims is subtracted from the justice value of the earnings terms, thus yielding a justice evaluation in justice units.

This formula has strengths and weaknesses. Its main merits are that (1) it translates all value or worth into justice units that are meaningful;

(2) it produces an unambiguous justice evaluation; that is, zero always means justice, a positive number always means unjust overreward, and a negative number always means unjust underreward; and (3) by incorporating a quadratic term for earnings, the formula produces a justice evaluation of greater absolute magnitude for underreward than for overreward.

But it is marred by several major difficulties. First, the best equation Jasso and Rossi were able to fit to the data was additive with a single nonlinearity in actual earnings, a nonlinearity they represented by inclusion of both a linear and a quadratic term for earnings. They did not investigate the possibility that the nonlinearity might be more faithfully represented in other ways.

Second, the Jasso and Rossi formula, by making the regression coefficients of paramount importance, is valid for only one spatiotemporal point. Without the regression coefficients, no prediction of the justice evaluation is possible. Since the slopes estimate the worth of particular claims to particular goods in a particular historical setting, they are ad hoc quantities which do not elucidate any general (theoretical) justice process.

Third, the Jasso and Rossi formula is made unduly complicated by its reliance on regression coefficients and quadratic terms, like that of Walster et al., by the use of ad hoc exponents; indeed, both are reminiscent of the blissful spawning of epicycles that characterized pre-Keplerian astronomy.

Now, if reality is faithfully approximated by their quadratic function, then we must swallow its complexity; if the data on earnings contain no underlying theoretical relation, then we must accept its particularity. But before reaching any conclusion, the data merit reanalysis.

In the remainder of the paper, I reanalyze the Jasso and Rossi data set and indeed find a spartan and general specification of the justice evaluation function and one related to the work of both Homans and Berger et al.

METHOD

A major task of this paper is to describe the empirical analyses that produced the new specification of the justice evaluation function. In this section I first describe the data set and then the mathematical analyses.

The Data

I reanalyzed a set of data collected and first analyzed by Jasso and Rossi (1977). They elicited from a probability sample of 200 respondents a set of almost 12,000 judgments about the justice or relative injustice of the earnings received by 600 distinctly different fictitious individuals and families. The judgments were obtained in household interviews by presenting to the respondents a set of descriptions of the fictitious units (termed "vignettes") and asking them to rate each description on a justice evaluation scale.

The respondent sample.—The respondent sample consisted of 200 white adults residing in private households in Baltimore City. Block quota sampling methods were used to choose the sample. Sex quotas were fixed to ensure that equal numbers of adult females and males were in the sample.

The vignette sample.—The vignette sample was constructed by means of a technique developed by Rossi and described in Rossi et al. (1974). Briefly, the fictitious units are generated by a random combination of preselected personal attributes, the major virtues of the procedure being the randomization of the experimental “stimuli” (i.e., the attributes) and the subsequent minimization of the problem of multicollinearity, and the principal shortcoming being the artificially platykurtic distributions of the variables.

The attributes used to describe the fictitious persons were as follows: (1) sex; (2) marital status, single or married; (3) number of children, zero to six, in increments of one; (4) education, years of formal education ranging from completion of seventh grade to college graduation, in increments of one year; (5) occupation, 96 occupational titles whose prestige scores, standardized by Bose (1973) for the Baltimore area, ranged from 8.0 to 95.8, plus the designation “unemployed and looking for a job” and the sex-restricted title “housewife”; (6) earnings, 15 unequally spaced levels of annual earnings ranging from \$2,000 to \$40,000 (thus, a married couple may have combined earnings ranging from \$2,000 to \$80,000); and (7) age held constant at 37 years for males and 35 years for females.

Half the vignettes describe one adult person designated as either married or single but provide no information about the married person’s spouse. These vignettes mirror some real-world situations, either of single persons living alone or of married persons about whose spouse one has no knowledge. The other half of the vignettes describe married couples, with all information given for both husband and wife. The vignettes were generated by computer and printed on IBM cards. Restrictions to eliminate nonsensical combinations (e.g., a physician with an eighth-grade education) were imposed a priori. To eliminate the possibility of zero earnings on a vignette, the unemployed and housewife designations were restricted to one such title per married couple.

Ten samples of 60 vignettes each were drawn, for a total of 600 distinctly different vignettes. The samples were drawn so that each contained five vignettes describing single males, five describing single females, 10 each of vignettes describing married males and married females (with no information about a spouse), and 30 vignettes describing married couples. The 10 vignette samples were systematically distributed with equal probability across respondents, each sample being given to 20 respondents, for a possible total of 12,000 judgments. To preclude order effects, each vignette set was well shuffled.

Inclusion of the unemployed and housewife designations resulted in 281

distinctly different two-earner married-couple vignettes and 19 one-earner married-couple vignettes.

The rating task.—Interviews lasting about half an hour were conducted by experienced professional interviewers in early 1974 in Baltimore. Each respondent was handed a set of vignettes and a box with nine slots, each labeled with a number and a description corresponding to a level of the justice evaluation variable, ranging from “−4 Extremely Underpaid,” through “0 Fairly Paid,” to “+4 Extremely Overpaid.” The task given to the respondent was to place each vignette in the slot that corresponded to the respondent’s justice evaluation of each vignette.² The interviewers reported that the respondents generally approached the rating task with seriousness, a view borne out by the number of ratings made, a total of 11,806 or over 98% of the possible ratings.

Data Analyses

The Jasso and Rossi (1977) research was so designed that it produced data pertinent to two major questions of distributive justice as regards earnings: (1) What is regarded as just earnings? (2) In the respondents’ view, how unjust are departures from just earnings? Indeed, answers to the first question are inferred from answers to the second. While in this paper we are mainly interested in the second question, that is, in specifying the justice evaluation function, the route to answering it takes us by the first question. I shall also present my own relevant findings.

Like Jasso and Rossi, I analyze each kind of vignette unit separately, for example, the set of single-person vignettes and the set of married-couple vignettes.

The original analyses, reported in Jasso and Rossi (1977), fitted quadratic regressions to the data. These were described as the best-fitting curves. The use of polynomial regression usually implies that the observed data curve resembles one or another of the polynomial curves of association (e.g., a parabola), and that the regression curve cannot be linearized by some transformation (e.g., root x). Now, scattergrams are essential for curve

² The actual wording of the item was as follows: “People and their jobs differ in a lot of ways. We have made up descriptions of different kinds of people and jobs. Here is a box with nine different slots in it. Here is a card with a description of a person and job. (a) Please put the card in the slot labelled Fairly Paid if you think that the salary is about right for the person, that is, if you think that the salary on the card is fair payment. (b) Put the card in the slot labelled Extremely Overpaid if you think that the money is way too much payment for these people, that is, if, taking everything into account, you think that they’re very much overpaid. (c) Put the card in the slot labelled Extremely Underpaid if you think that the money is way too little payment for these people, that is, if taking everything into account, you think that they’re very much underpaid. (d) If you think that the salary is a bit too much or a bit too little, that is, if you think that it belongs somewhere between the most overpaid and the most underpaid, then just put the card in the slot that you think matches the fairness or unfairness of the salary.”

fitting (see, e.g., Snedecor and Cochran 1967, pp. 447–71), just as they are for valid interpretation of the arithmetical results of the usual regression computations (see, e.g., Glass and Stanley 1970, p. 127). Since Jasso and Rossi reported no scattergrams, I decided to begin my reanalysis with close scrutiny of the original data. I plotted the observed justice evaluations on the actual earnings for the separate subsets of vignettes. As expected, plots of the raw observations produced relatively uninformative rectangular scatters. However, plots of the mean, median, and modal values of the observed justice evaluations produced curves of a very definite nonlinear shape. For example, a plot of the mean justice evaluation on the 51 observed values of combined earnings for the two-earner married couples produced a curve with a rather steep ascent to the vicinity of zero (the point of fair earnings) at around \$20,000 on the X -axis, followed by a long, slow climb to 2.949 at \$58,000, next a dip to 1.718 at \$68,000, and finally a recovery to 2.525 at \$74,000. Because the N 's for each mean (of the 51 plotted means) ranged from 17 to 279, there was some uneasiness about the results. But a plot of the mean justice evaluation of each of the 281 distinctly different two-earner married-couple vignettes (the N 's for each of the 281 plotted means range from 16 to 20) on the combined earnings again resulted in a peak mean fairness rating at \$58,000 and, more important, a similarly shaped curve. The curve itself provides some useful information: dollar increments of fixed size appear to “buy” more underpayment than overpayment.

These curves, plotted for all the subsets of the data, look amenable to representation either by a logarithmic curve or by a second-order polynomial (see Snedecor and Cochran 1967; Guilford 1954). Trying both, I found that fitting a parabola results in substantial distortion, for it is incapable of representing the postpeak recoveries in the justice evaluation that are observed in the data set (as in the case described above).

However, a logarithmic transformation of earnings produces curves that are both highly linearized and of unusual fidelity. Plots of both the mean justice evaluation for the natural logarithm of each value of earnings (for example, in the two-earner married-couple case, $N = 51$) and of the mean justice evaluation for the logarithm of earnings for each distinctly different vignette (e.g., in the same case, $N = 281$) indicate unmistakably that the justice evaluation is linearly related to the logarithm of earnings.

Thus, the plots of the original data furnished one important clue in my search for the “true” underlying justice evaluation function: the justice evaluation is a linear function of the logarithm of actual earnings. But unless all other things are equal, the logarithm of actual earnings is not sufficient to predict justice evaluations. So the search for the rest of the function continued.

Jasso and Rossi (1977) reported statistically significant terms for all the

other vignette attributes. So my next step was to regress the observed justice evaluations on all vignette attributes, with earnings represented solely by their natural logarithm (a more economical model than the two earnings terms required in the quadratic regressions). All quantitative variables were coded in their own units; sex and marital status entered the regressions as dummy variables, with the categories “single” and “male” coded zero; earnings were coded in thousands of dollars.

Table 1 reports the results of these regressions. The coefficients of multiple correlation are extremely close to those for the quadratic regressions (e.g., in the case of all one-person vignettes, the quadratic R^2 was .6616, while the new R^2 is .663), again confirming the view that whatever nonlinearities were in the data and were tapped by the quadratic term are faithfully approximated by the logarithmic transformation of earnings.³ Moreover, the plots show that the logged equations are more faithful by far than the quadratic ones.

Many additional models were tried on the data, for example, a variety of interaction terms both with and without the log of earnings. No set of interaction terms increased the coefficients of multiple correlation by as much as .01. The equations reported in table 1 represent the absolutely best-fitting curves to the data set.

Now, we think these equations are most faithful, yet we are still seemingly confined to prediction by a set of structural coefficients. If we wish to predict the justice evaluation for a given person of given attributes and given earnings, we literally have to take each attribute, multiply it by its b weight, and add the products—a disappointment.

Happily, that initial impression turns out to be wrong. It is not necessary to take a sum of products on a string of attributes in order to predict a justice evaluation. Using algebra, we can restate the same justice evaluation functions reported in table 1 in a variety of equivalent expressions, finding at last the truly superior justice evaluation function—the candidate I am offering for the Law of Justice Evaluations.

In order to make the sequence of equivalent expressions as clear as possible, we begin with what may appear to be a sidetrack but in fact is not, the estimation of just earnings functions. If we wished to employ the equations reported in table 1 to estimate just earnings for particular earners, we would do the following: set the justice evaluation at zero (the point of perfect justice on the justice evaluation scale) and solve for earnings.

We now derive the underlying justice evaluation function using the data in the single-person equation. The same can, of course, be done for all the

³ Regressions of the observed mean justice evaluation at the level of the vignette, rather than of the raw observation at the level of the individual rating, produce R^2 s close to unity. E.g., in the case of the one-earner married couples, at vignette level ($N = 19$), $R^2 = .9718$, while at rating level ($N = 379$), $R^2 = .710$.

TABLE 1
ESTIMATED JUSTICE EVALUATION FUNCTIONS

Vignettes	Justice Evaluation =	R ²	N
One-person-single.....	−3.784 + 2.2364 ln earns. −0.0779 ed. −0.0154 occ. + 0.3469 sex	.685	1,994
One-person-married.....	−3.830 + 2.0949 ln earns. −0.0423 ed. −0.0192 occ. + 0.1887 sex −0.0748 ch.	.651	3,986
All one-person.....	−3.661 + 2.1389 ln earns. −0.0556 ed. −0.0181 occ. + 0.2440 sex −0.0756 ch. −0.1827 MS	.663	5,980
Two-earner married-couple.....	−5.554 + 2.4691 ln comb. earns. −0.0378 HEd −0.0269 WEd −0.0158 HOcc −0.0097 WOcc −0.0424 ch.	.501	5,447
One-earner married-couple.....	−4.478 + 2.262 ln earns. −0.0582* ed. −0.0135* occ. + 0.3696* sex −0.0204 ch. ^a	.710	379

NOTE.—All R² and b-weight values are statistically significant beyond the .001 level except as noted below. ch. = children; MS = marital status; HEd = husband's education; WEd = wife's education; HOcc = husband's occupation; WOcc = wife's occupation.

^a N.S.

* P < .05.

other equations. We begin by deriving the just earnings function.

$$\ln \text{ just earnings} = \frac{3.784 + 0.0779 \text{ ed.} + 0.0154 \text{ occ.} - 0.3469 \text{ sex}}{2.2364}$$

Next, restating the original equation,

$$\begin{aligned} \text{justice evaluation} &= 2.2364 \ln \text{ earnings} - 2.2364 \\ &\quad \times \left[\frac{3.784 + 0.0779 \text{ ed.} + 0.0154 \text{ occ.} - 0.3469 \text{ sex}}{2.2364} \right] \end{aligned}$$

Hence:

$$\text{justice evaluation} = 2.2364 \ln \text{ actual earnings} - 2.2364 \ln \text{ just earnings}$$

it then follows that

$$\text{justice evaluation} = 2.2364 (\ln \text{ actual earnings} - \ln \text{ just earnings})$$

and that

$$\text{justice evaluation} = 2.2364 \left[\ln \frac{\text{actual earnings}}{\text{just earnings}} \right]$$

This means that the justice evaluation is a logarithmic function of the ratio of actual earnings to just earnings or, alternatively, a function of the signed difference between the logarithms of actual earnings and of just earnings. This formula produces a justice evaluation in meaningful justice units, and it quantifies the human experience that deficiency is felt more keenly than excess. The multiplicative constant transforms the raw justice evaluation into the units of the particular justice scale used. For example, in the present case, table 1 shows that the constant hovers about 2.2 and varies only slightly for the two-earner married couples.

In the next section I shall discuss in detail the form of the new justice evaluation function and its implications. For the moment, let us review what we have done empirically to reach it. First, we fit the best and most faithful representation of the data, and, second, we found within the equation of that best-fitting curve a new and more general specification of the justice evaluation function, one that may turn out to be a universal Law.⁴

⁴ The justice evaluation function provides a good illustration of Blalock's (1971) remarks on the inappropriateness of standardized regression coefficients for theory construction. E.g., if the justice evaluation function were operating, i.e., if it were in effect a law, then empirical regression analysis would always yield unstandardized coefficients for actual earnings and just earnings of about 1 and -1, respectively; but the beta weights would fluctuate wildly with the observed variances of the two predictor terms, variances which may differ in fact across populations or which may differ artificially in experimental settings. Like the betas, the notions of variance decomposition and of variance explained by

As noted earlier, finding the best-fitting curve also produces information on the question of the determinants of just earnings. In our derivation of the justice evaluation function we expressed the equations reported in table 1 as estimates of just earnings. Now, by restating those equations, we can derive a new just earnings function. Continuing with the equation we have been using for illustrative purposes, that for the one-person-single vignettes, we derive the just earnings function:

$$\ln \text{ just earnings} = \frac{3.784}{2.2364} + \frac{0.0779 \text{ ed.}}{2.2364} + \frac{0.0154 \text{ occ.}}{2.2364} - \frac{0.3469 \text{ sex}}{2.2364},$$

$$\ln \text{ just earnings} = 1.692 + 0.0348 \text{ ed.} + 0.0069 \text{ occ.} - 0.1551 \text{ sex},$$

$$\text{just earnings} = (5.4303)(1.0354^{\text{ed.}})(1.0069^{\text{occ.}})(0.8563^{\text{sex}}).$$

Table 2 reports the final just earnings functions for all subsets of the vignettes. It is immediately clear that there are important commonalities between all these equations and that they have an intuitively clear meaning. They are exponential functions in which the attributes of the earner produce returns calculated on a principal amount.

These just earnings functions represent faithfully the relations in the data set. But they do not have the same status as our candidate for Law of Justice Evaluation.⁵ We cannot claim universality for them. We cannot claim that the respondents would recognize the procedure. It is possible that the respondents are using some other measure of just earnings, say, the mean or the median of earner classifications, a measure that may be coincidentally correlated with the predictions made by the just earnings functions.

On the other hand, we note briefly the possibility that these just earnings functions mirror a process in which the principal represents basic or minimum earnings for all earners (perhaps even a minimum allowance for all adults) based on the principle of need and that the other terms represent just re-

each of the two independent terms are inappropriate and misleading. On the other hand, while the betas are worthless in the task of confirming general laws, they do provide one highly important piece of information about observed conditions in the research setting (natural or artificial). The term with the largest beta has the largest variance. Hence, the betas indicate unmistakably what sort of dispersion is deemed just in a particular population. E.g., in the data analyzed here, actual earnings "account for about 90% of the explained variation in the justice evaluation" and just earnings (understood as the set of justly perceived claims on earnings) "account for the remainder"; hence, we can infer that the acceptable variation in just earnings is considerably smaller than that put experimentally into actual earnings. Note that this illuminates our understanding of "what people think is just," while remaining neutral to the task of specifying the justice evaluation function.

⁵ I am grateful to an anonymous referee for emphatically pointing this out.

TABLE 2
ESTIMATED JUST EARNINGS FUNCTIONS

Vignettes	Just Earnings =	R ²	N
One-person-single.....	(5.4303)(1.0354 ^{ed.})(1.0069 ^{occ.})(0.8563 ^{sex})	.685	1,994
One-person-married.....	(6.2227)(1.0204 ^{ed.})(1.0092 ^{occ.})(0.9138 ^{sex})(1.0363 ^{ch.})	.651	3,986
All one-person.....	(5.5378)(1.0263 ^{ed.})(1.0085 ^{occ.})(0.8922 ^{sex})(1.0359 ^{ch.})(1.0892 ^{MS})	.663	5,980
Two-earner married-couple*.....	(9.4821)(1.0154 ^{HEd.})(1.0110 ^{WEd.})(1.0064 ^{HOcc.})(1.0039 ^{WOcc.})(1.0173 ^{ch.})	.501	5,447 ²
One-earner married-couple.....	(7.4745)(1.0265 ^{ed.})(1.0061 ^{occ.})(0.8470 ^{sex})(1.0092 ^{ch.})	.710	379

NOTE.—See notes to table 1.
* Just combined earnings.

turns based on the principle of desert. I shall not comment further on the just earnings functions in this paper.

In the next section, I discuss the justice evaluation function.

THE NEW JUSTICE EVALUATION FUNCTION

The Form of the Function

The present data indicate that the justice evaluation of earnings is a logarithmic function of the ratio of actual earnings to just earnings:

$$\text{justice evaluation} = \ln \frac{\text{actual earnings}}{\text{just earnings}}.$$

In other words, when an observer judges the fairness or unfairness of an individual's earnings, the mental operation involves a ratio procedure. A mental image of what just earnings would be for such a person is quickly retrieved or formed, a ratio made of the actual earnings to the just amount, and "something" that can be formally represented by a logarithm taken of the ratio.⁶

The justice evaluation score associated with an individual who earns exactly his or her just earnings would be zero, which is the logarithm of the ratio one. Zero represents the point of justice, and such a person would be considered justly rewarded. When actual earnings exceeds just earnings, the ratio is larger than unity and its logarithm is a positive number, indicating unjust overreward. When just earning exceeds actual earnings, the ratio is a proper fraction and its logarithm is a negative number, indicating unjust underreward.

This means that the magnitude of injustice is equal for any fraction of one and for its reciprocal, because the logarithm of such a fraction and of its reciprocal are identical except in sign. To illustrate: the magnitude of injustice associated with an individual who earns one-half of his or her just earnings (whatever that amount may be) is -0.6931 (rounded), which is the natural logarithm of the ratio 0.5. The justice evaluation score of -0.6931 indicates that the individual is unjustly underrewarded and that the injustice is of a magnitude of 0.6931. The justice evaluation score of an individual who earns twice his or her just earnings is $+0.6931$ (rounded),

⁶ The form of the justice evaluation function is similar to that of Bernoulli's utility function and of Weber-Fechner's sensation function (Stevens 1975). However, in both the utility and the sensation functions, the numerator of the ratio is larger by definition than the denominator (in the case of utility, the numerator contains an increment to the denominator; in the case of sensation, the numerator is a stimulus greater than a threshold value placed in the denominator), thereby producing functions that range only over the positive segment of the real-number line. To my knowledge, the justice evaluation function is the first logarithmic specification proposed in the sociobehavioral sciences in which two terms are freely varying and whose predicted values span the full real-number line.

which is the logarithm of the ratio 2. This score of $+0.6931$ indicates that the individual is unjustly overrewarded and that the magnitude of the injustice is 0.6931. Similarly, the magnitude of the injustice associated with an individual who earns one-third of his or her just earnings would equal that associated with an individual who earns three times his or her just earnings, and so on. For example, if just earnings are \$10,000, then the absolute magnitude of injustice is 0.6931 equally for actual earnings amounts of \$5,000 and of \$20,000; in this case, a deficiency of \$5,000 produces the same magnitude of injustice as an excess of \$10,000.

These properties of the justice evaluation function may be summarized by saying that the geometric progression of the ratio of actual earnings to just earnings produces an arithmetic progression in the corresponding sequence of justice evaluation scores and that the common difference in the sequence of justice scores is the logarithm of the common ratio of the geometric progression.

Thus, to produce an increment of x in the justice evaluation score, a given ratio of actual to just earnings would have to be multiplied by the antilogarithm of x ; to produce a decrement of x in the justice evaluation score, a given ratio would have to be divided by the antilogarithm of x .

This means that, holding just earnings constant, to produce an increment of size x in the justice evaluation, it is necessary to increase actual earnings by adding to it an amount equal to $(e^x - 1)$ of actual earnings; to produce a decrement of size x in the justice evaluation, it is necessary to decrease actual earnings by subtracting an amount equal to $(1 - e^{-x})$ of actual earnings.

Now, for all positive x , the relative decrement $(1 - e^{-x})$ is smaller than the relative increment $(e^x - 1)$. Formally,

$$[(1 - e^{-x}) < (e^x - 1)], \quad x > 0.$$

This inequality quantifies the common human experience that deficiency is felt more keenly than excess.

The ratio of the relative decrement $(1 - e^{-x})$ to the relative increment $(e^x - 1)$ is $(1/e^x)$. This ratio indicates that the dollar amount of the deficiency is related to the dollar amount of the excess as the reciprocal of the antilogarithm of the absolute value of the justice evaluation. Thus, in the case above where, given just earnings of \$10,000, actual earnings of \$20,000 and of \$5,000 produced justice evaluations of 0.6931 and -0.6931 , respectively, the deficiency of \$5,000 is $(1/e^{0.6931})$ or one-half the excess amount of \$10,000.

Finally, we note that the logarithms of the ratios of actual earnings to just earnings may be regarded as "raw" justice scores. Multiplication by a constant transforms them to justice scores on particular justice scales. For example, a constant of about 2.2 transforms the raw scores to the nine-point

justice scale used in the present research. But, whatever the transformation, the relation remains invariant: it is judged equally unfair to earn one-half or two times one's just earnings.

The Justice Evaluation Function in Classical Perspective

Let us see now how the new specification fares when placed in the classical contexts of social science discourse. We return to the models proposed by Homans (1974, 1976) and by Berger et al. (1972) discussed earlier.

The Homans model.—With a little nimbleness, we can equate Homans's reward term with the actual earnings term and his investment term with the just earnings term (for just earnings is a function of the various claims on earnings allocation, a claim is akin to an investment, and, of course, both terms must be expressed in the same currency) and produce a ratio that Homans (or Aristotle) might recognize. Now, if these ratios are equal for two earners, Homans would say that justice obtains in the system. This is a perfect consequence of his position that "whatever is, is right," as long as whatever is, is consistently and impartially so. Our formulation, on the other hand, would produce equal justice scores, but not necessarily justice. Equal justice scores entail no specific condition of justice; they may be both -3 or both zero; that is, they may be both excruciatingly unjust or both perfectly just.

But now look at the present formula a little closer. If an observer indeed judges that "whatever is, is right," that, say, whatever a worker manages to get on the market is fair, then such an observer makes justice evaluations in this manner: he or she looks at the numerator, places that same amount in the denominator, and the result is perfect justice. This is a useful feature for observers who regard injustice primarily as underpayment and who have difficulty with the notion of "unjustly too much." Our formula, then, has the virtue of providing an unambiguous measure of the justice evaluation, even under conditions of idiosyncratic rules of just allocation. Hence, the investigator needs to know only an observer's standards of justice for any earner (or any category of earner) to predict his or her justice evaluation for any earner. Of course, consensually held standards of justice are much more powerful, for then the investigator does not need to ascertain an observer's standards of justice in order to predict his or her justice evaluations.

The model of Berger et al.—Our new specification is an accurate, but more extensive and precise, representation of the model proposed by Berger et al.; it uses the same two terms, for the real-world actor's level of a goal-object can be construed as actual earnings, and the "referential structure" level of a goal-object can be construed as just earnings. Our formula takes these terms and specifies the nature of the comparison as one not of arithmetic difference but, more precisely, of proportion (or the difference between two

logarithms). Our model leads to a statement of the precise sort of injustice, that is, of deficiency or of excess, and it predicts the exact magnitude of injustice.

The models of Homans and Berger et al.—Examination of the two classical models in light of the justice evaluation function sharpens them, enabling us to see clearly that the major distinction between them lies in the etiology of the just earnings term. We think that both Homans and Berger and his associates would agree that our formula accurately quantifies their thinking on justice evaluations and on types and degrees of injustice; Homans would no doubt feel comfortable with the logarithm of a ratio, and Berger et al. would no doubt feel comfortable with the difference between two logarithms. The disagreement between them seems to be exclusively about just shares, about whether the principles of justice actually held by individuals and societies are based on existential or utopian criteria. Put simply, their argument is not about how people judge relative injustices after they have formed notions of justice but, rather, about how they first acquire those notions of justice.

Generalization of the Justice Evaluation Function

Empirical support for our specification of the justice evaluation function has been reported for the case of earned income. I now propose its extension to cover all socially distributed goods. The most general statement is

$$\text{justice evaluation} = \ln \frac{\text{actual amount of good}}{\text{just amount of good}}.$$

Stated in this manner, it can be applied to whatever good one pleases, to land, livestock, inheritance, children's allowances, hours of overtime work, incentive pay, and so on.

It can readily be seen that the justice evaluation function, applied to any good of distribution, always produces precise and unambiguous evaluations in meaningful justice units. It is, therefore, a strong candidate for a universal Law of Justice Evaluation.

SUMMARY AND A LOOK AHEAD

In this paper I have reported the empirical induction of a new specification of the justice evaluation function. It has been shown to be well supported for the case of earnings. I have proposed it as a universal Law of Justice Evaluation about shares of all goods of social distribution.

Let us see now the place such a Law occupies in the scientific study of the sense of justice and injustice. We consider that a theory of distributive justice must contain at least three Laws, that is, that it must perform three

tasks. First, it must specify determination of the “just” term, of what is actually considered just by individuals and societies. Second, it must specify the kind and degree of injustice associated with given departures from justice, that is, it must specify the justice evaluation. Third, it must specify behavioral consequences of the justice evaluation.

I submit that we have accomplished the second of the tasks requisite to a theory of distributive justice. Work must now proceed on the other two.

With respect to the first, I believe that the lines of the argument have been well drawn by Homans and Berger et al.: Are persons’ notions of “what is just” formed by reference to “what is” or to “what ought to be”? In the course of daily life, I have observed the operation of both existential and utopian criteria, and so I suggest that the question may be more fruitfully posed as follows: What accounts for the observed variability in individuals’ formation of the just term?⁷

The third task involves determination of the following: (1) both individual and social responses (2) to both kinds of injustice (3) judged both about oneself and about others. The third task in the construction of a theory of distributive justice is perhaps the most difficult. But many investigators have been gathering data in many settings—in the historical record, in the laboratory, and in contemporary society. Correlation analysis of observed behaviors and observed justice evaluations may soon make possible explanation and prediction of many personal and social phenomena.

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⁷ I consider that much of the dynamics of social life centers on determination of the “just” term and that the force of ideology is here particularly visible. A practical implication of this view is that the strategy of choice for revolutionary propagandists is not persuasion of the citizenry that a given actual condition is unjust but, rather, persuasion of the citizenry that some other, nonfactual, very precisely described condition is just. For once the citizenry accepts as just some envisioned state of things, the judgment that the actual condition is unjust follows inexorably.

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